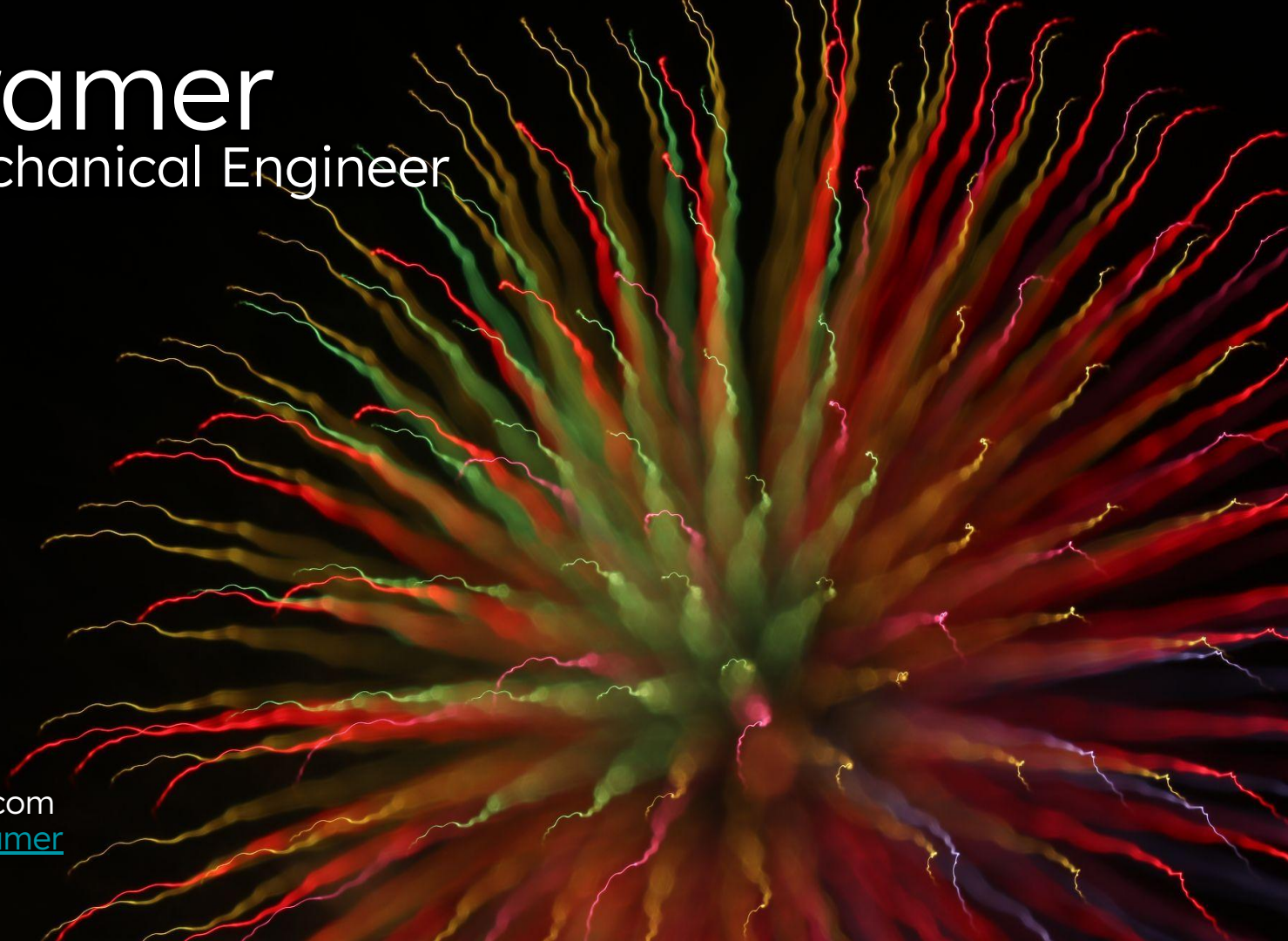


Erik Kramer

Robotics Mechanical Engineer
Portfolio

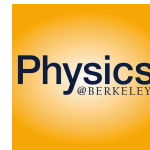
erik.h.kramer@gmail.com
[linkedin.com/in/ehkramer](https://www.linkedin.com/in/ehkramer)
(310) 780-7076



Erik Kramer | About Me

I am a hardware-first, hands-on roboticist with a strong mechanical foundation, specializing in taking complex real-world physical robotic systems and mechanisms from concept to fielded hardware. I bring a cradle-to-grave engineering mindset grounded in classical mechanical engineering and strengthened through experience across robotics, human-robot interaction, and technology development for space systems. My background bridging engineering design and experimental physics allows me to approach complex problems with the executional rigor of an engineer and the curiosity-driven perspective of a scientist.

Experience



Education



Ph.D. Mechanical Engineering
M.S. Mechanical Engineering



B.A. Physics
B.S. Mechanical Engineering

Erik Kramer | Major Projects List

Robotics, Mechanisms & Integrated Systems

- ❖ Mobile ISAM Robotic Arm Testbed (SAMMI)
- ❖ Hazardous Environment Rovers
- ❖ Virtual Reality Exoskeleton (V-Rex)
- ❖ Ocean World Lander Autonomy Testbed (OWLAT)
- ❖ Europa Sample Return Mission Concept Study
- ❖ SuperCDMS SNOLAB Tower Standoffs
- ❖ Assistive Gripper Mechanism
- ❖ Compliant Micro Mirror Mechanism

Scientific Research & Analysis

- ❖ Airborne Optical/SWIR Imagery (SAR-Fusion)
- ❖ Lower Limb Exoskeleton Safety Analysis
- ❖ Cryogenic Properties of Engineering Materials
- ❖ Microfluidic Viscometer

Space Technology & Flight Projects

- ❖ In-Space Assembly Technology Development
 - Autonomous Surface Solar Array Assembly
 - Robotic Satellite Servicing
 - Multi-Rover Lunar Surface Logistics
 - Segmented Telescope Robotic Assembly
- ❖ Robotic Lunar Regolith Sampling Test
- ❖ Mars Helicopter Motor Verification Test System

Just for Fun

- ❖ Mars 2020 Remote-control Rover Backpack
- ❖ Robotic Knife Game
- ❖ Comic Convention Costume Design
- ❖ NASA JPL Pumpkin Carving Contest
- ❖ Wall Perching Quadcopter

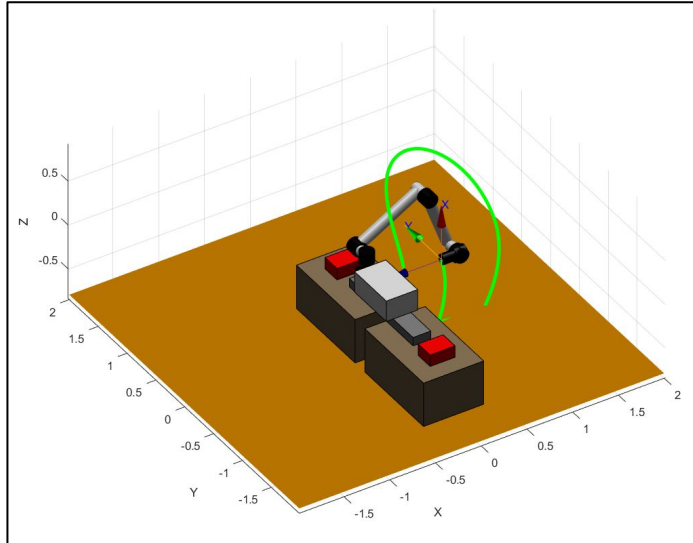
Interested in one of my other big projects not shown in the following slides?

→ *Let's chat about it!*

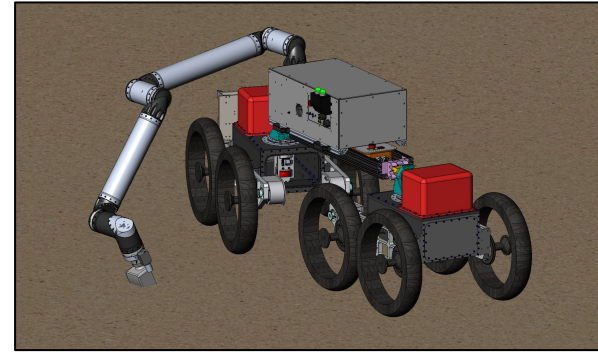
Erik Kramer | Mobile ISAM Robotic Arm Testbed (SAMMI)

Project → An untethered reconfigurable robotic testbed for in-space servicing assembly and manufacturing development

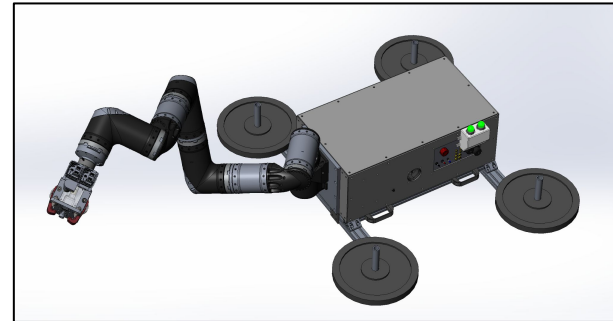
Tasks → Defining requirements, **arm design** and **analysis**, avionics **housing** design, **control software** development



Kinematic Analysis Driven Arm Design



Modular Design Approach



Erik Kramer | Mobile ISAM Robotic Arm Testbed (SAMMI)

Objectives

- Build an ISAM robotic arm testbed capable of digging and manipulation when stable or on a mobile base
- Design a new digging arm that can effectively collect soil while mounted atop a robotic rover base
- Create a base to house avionics and mechanically support different arms

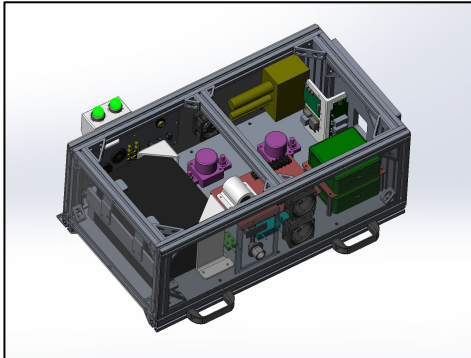
Process

- Utilized **MATLAB** to model arm candidates and select a design
- Employed **SolidWorks** to design new components to interface with existing heritage hardware
- Designed **housing** for electronics and deployed in-house software for **arm control**

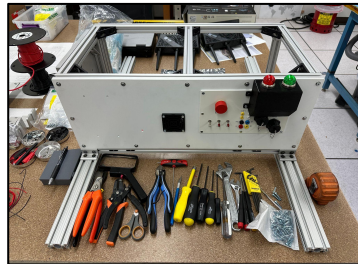
Results

- Produced two physical testbeds that can be configured with different arm types and can be used as a stand alone system or on a mobile base
- Utilized Testbed in ISAM demos including satellite servicing, segmented telescope assembly, and lunar solar panel assembly

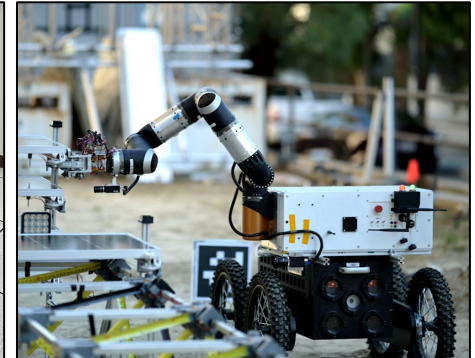
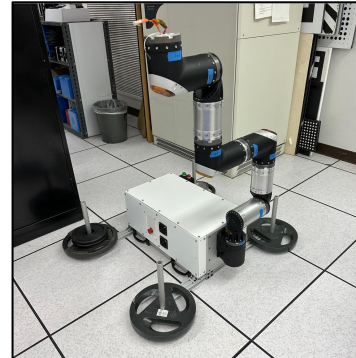
Harness Routing Aware Avionics Configuration



Electromechanical Build



System Bring-up and Testbed Operation

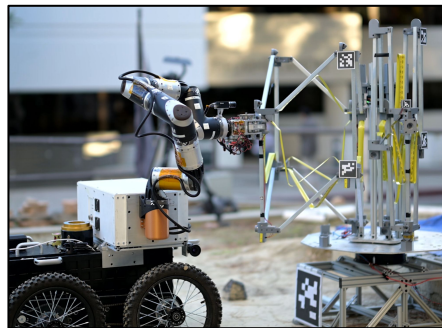
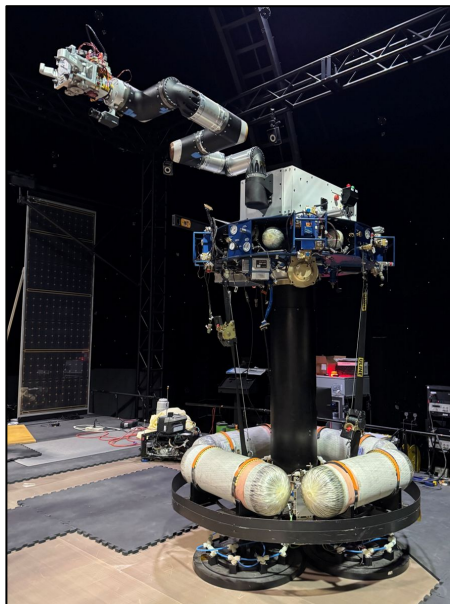
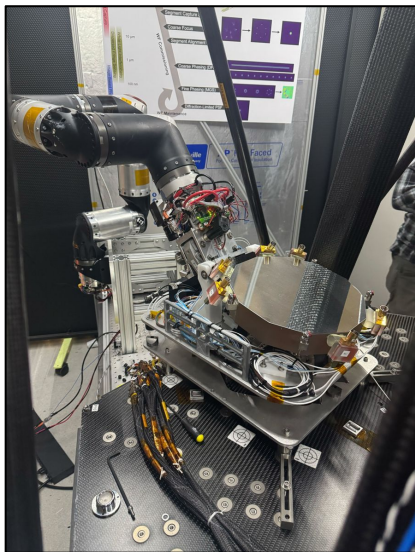


Erik Kramer | In-Space Assembly Technology Development

Project → Various technology demonstrations of autonomous in-space assembly related tasks using robotic arms

Tasks → **End-to-end** technical **leadership** for **conops**, hardware **integration**, **software**, and **autonomy** operations

Integration for In-Orbit Simulation



Surface Mobile Manipulation Autonomy Development

Erik Kramer | In-Space Assembly Technology Development

Objectives

- Develop and demonstrate in-space servicing, assembly and manufacturing technologies at TRL levels 3 to 5

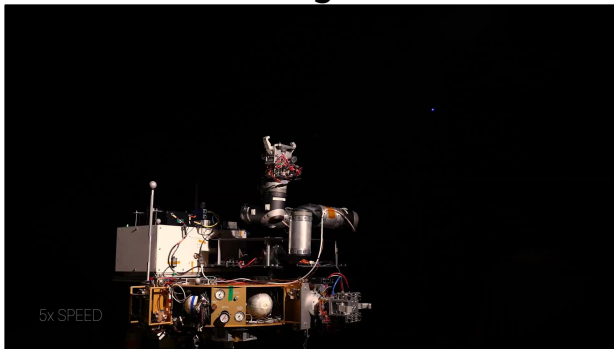
Process

- Led **technical efforts** on a cross disciplinary team and maintained full systems level understanding
- Utilized **vision** and **force control** for manipulation and mobility systems
- Problem solved hardware and software issues on **aggressive timelines** to ensure successful demos

Results

- Demonstrated autonomous technologies for satellite servicing, segmented telescope assembly, surface solar array assembly, and multi-rover Lunar surface logistics

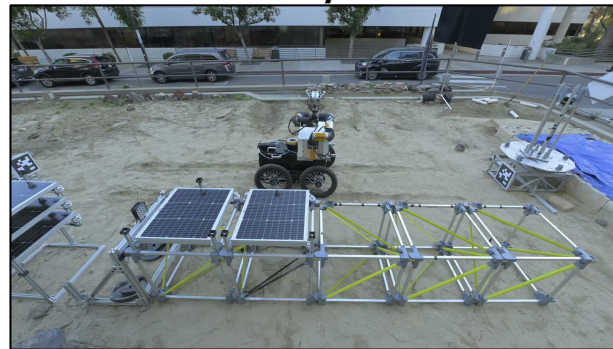
[Click](#) for Satellite Servicing Demo



[Click](#) for Deployable Truss Structure Assembly Demo



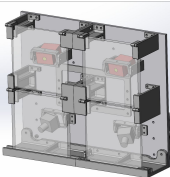
[Click](#) for Solar Array Assembly Demo



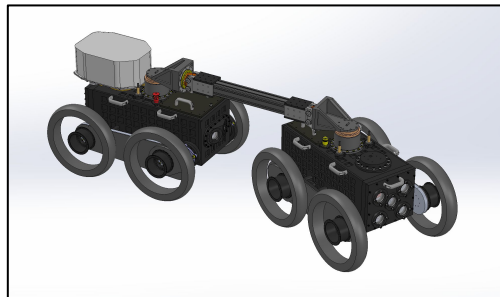
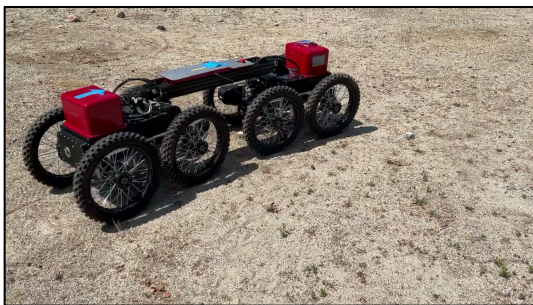
Erik Kramer | Hazardous Environment Rovers

Project → Dual body robots capable of traversing hazardous environments and difficult terrain

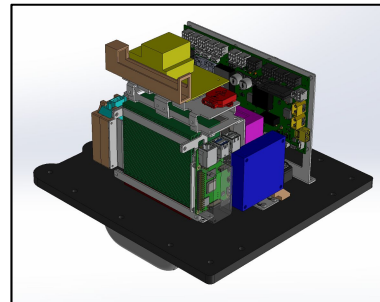
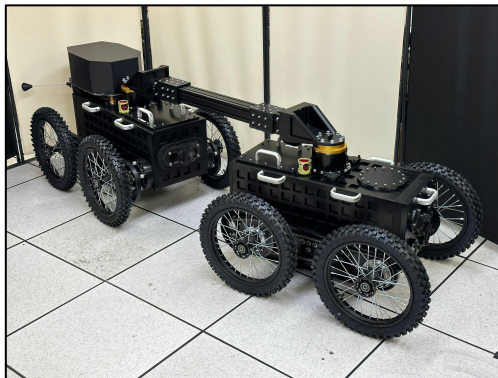
Tasks → **Structure design** and **delivery**, electronics **packaging**, integration and test, next-gen **mobility** development lead



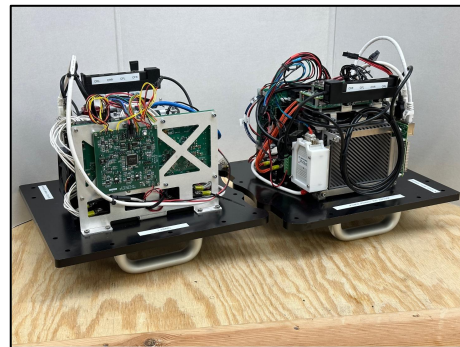
Rapid Prototype Development and Mobility Testing - [Click](#) for Video



Hardware Delivery And Subsystem Integration



Configuration Management and Electronics Packaging



Erik Kramer | Hazardous Environment Rovers

Objectives

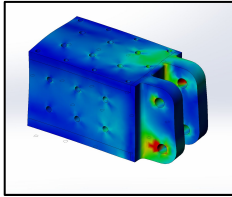
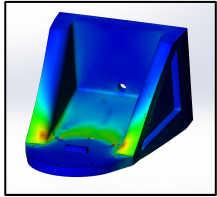
- Create a fieldable explosion-proof robotic system capable of traversing hazards in tight spaces in mines
- Design and deliver the central structure, electronics tower hardware, and next-gen steering actuators
- Work with a team to integrate and field test rover systems

Process

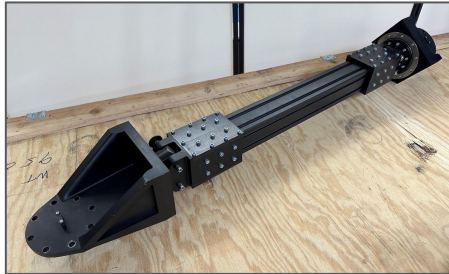
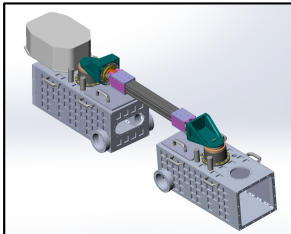
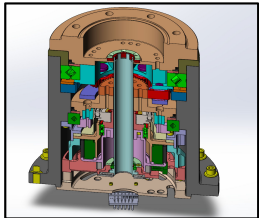
- Filled **responsible engineer** role for central structure and mobility subsystems, leading CAD, analysis, procurement, integration, and test
- Led **design reviews** to ensure subsystem requirements were met
- Provided **code**, **analysis**, and **hardware** to other subsystems

Results

- On time integration of hardware for delivery of completed 2nd gen robot to stakeholders
- Completed detailed documentation for actuator load analysis and field performance validation results
- Implemented and tested driving algorithms in the field



Design and Analysis



Parts Procurement and Assembly



[Click Here for Field Test and Autonomy Videos](#)

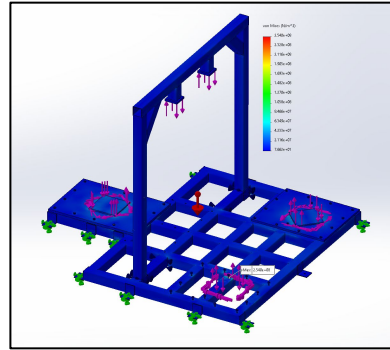
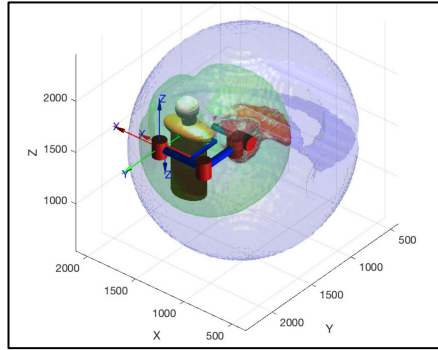
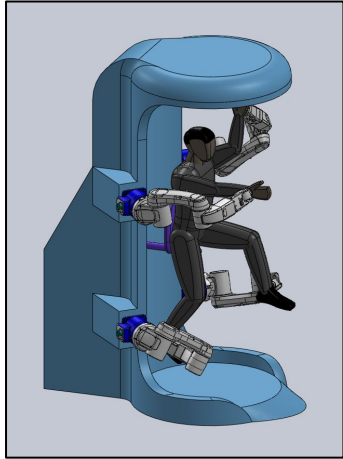


Erik Kramer | Virtual Reality Exoskeleton (V-Rex)

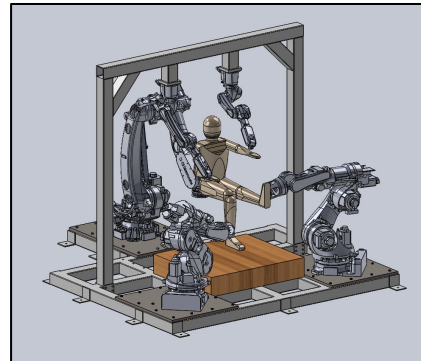
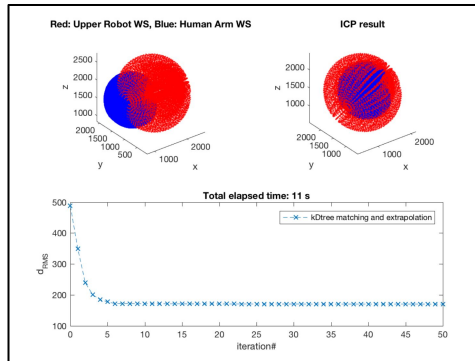
Project → A five robotic arm system that emulates virtual reality environments to a user through haptic feedback

Tasks → Ownership of the **end-to-end development** for the entire **hardware system** and software **controller**

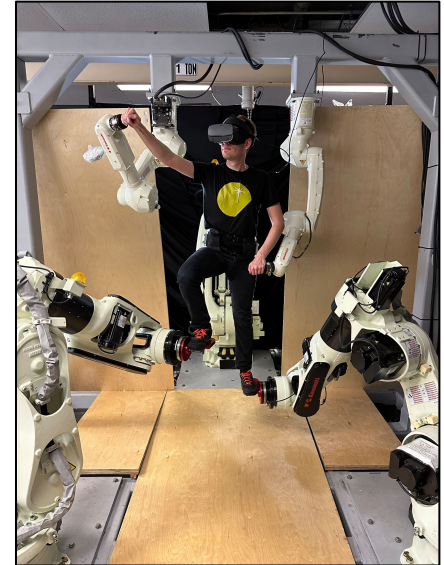
Conceptualization



Analysis Driven Iterative Design



Product Delivery



Erik Kramer | Virtual Reality Exoskeleton (V-Rex)

Objectives

- Design and fabricate a novel full body haptic device integrating industrial robots and custom parts
- Develop a safety-conscious controller to allow for human-robot interaction through force feedback

Process

- **Cognizant management** of all aspects of development, fabrication, assembly, and integration
- Led support engineers and **reviewed** their work and drawings
- Created hardware designs, configurations, and drawings
- Developed **controller architecture**

Results

- Demonstrated design and functionality of real time safety features through experimental data
- Created a system that allows a user to feel a virtual environment
- Published methods and design in peer reviewed robotics journal

[Click](#) for Safety Control Feature Video Demo

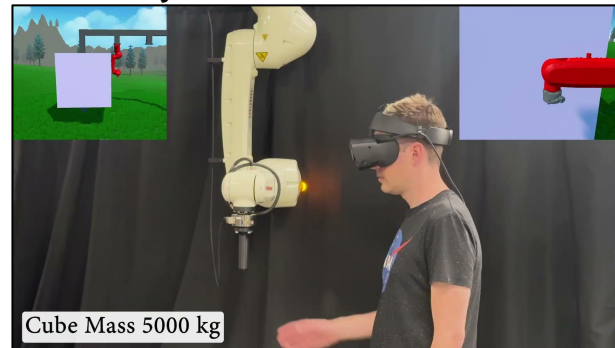
Safety-Focused Admittance Control for Physical Human-Robot Interaction with Rigid Multi-Arm Serial Link Exoskeletons

Bionics Lab
University of California, Los Angeles (UCLA)

[Click](#) for Full Body Virtual Reality Demo



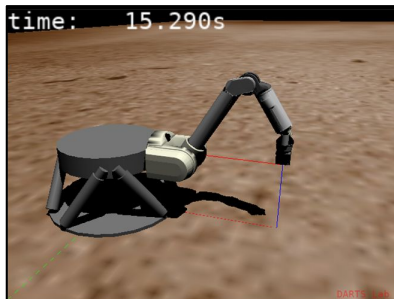
[Click](#) for Haptic Object Interaction Demo



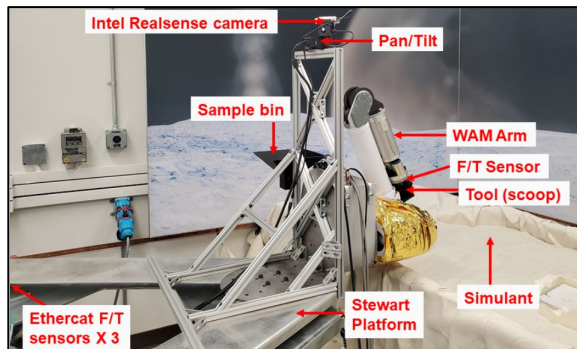
Erik Kramer | Ocean World Lander Autonomy Testbed

Project → A lander and robot arm sampling testbed to evaluate the performance of user autonomy algorithms

Tasks → Non-earth gravity **dynamics** emulation, **motion planning** algorithms, development of user features/**sequences**



Software Simulation and Hardware Operation



[Click](#) for Low-Gravity Emulation Video Demo

This research was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration (80NM0018D0004)

Physical Emulation of Low-Gravity Ocean World Lander-Manipulator Dynamics on a Robotic Testbed

Erik Kramer, Ashish Goel, Anna Boettcher,
Erica Tevere, and Hari Nayar



Jet Propulsion Laboratory
California Institute of Technology

© 2024 California Institute of technology. Government sponsorship acknowledged

Erik Kramer | Ocean World Lander Autonomy Testbed

Objectives

- Develop a mode to emulate non-earth gravity dynamics through torque offloading with software
- Solve kinematics issues causing Cartesian motion planner to find bad trajectories that result in faults
- Add user features and sequences as needed by autonomy teams

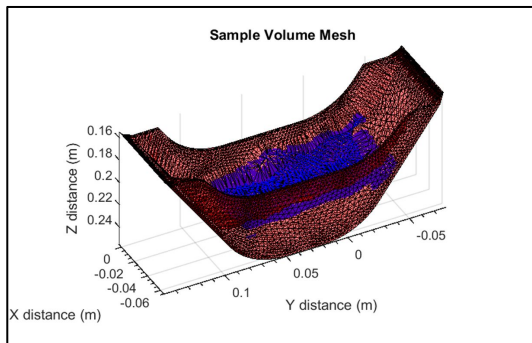
Process

- Analyzed telemetry with MATLAB to find root cause of robot faults
- Wrote new and updated old C++ code in a multi-author repository
- Utilized MATLAB, in house simulator, and hardware testing to verify methods

Results

- Demonstrated non-earth gravity dynamics through torque control
- Implemented a new motion planning optimizer that finds smooth trajectories and is 30% faster
- Added sequences such as radial scooping and features such a ROS interface for an autonomy subsystem

Testbed Feature Development



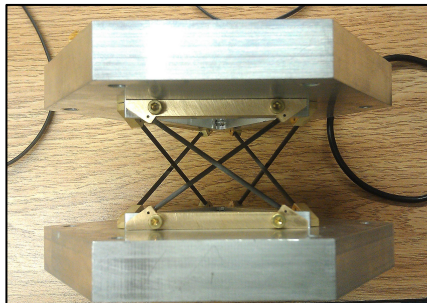
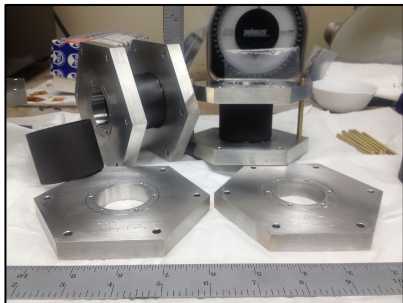
[Click](#) for Scoop Sequencing Video



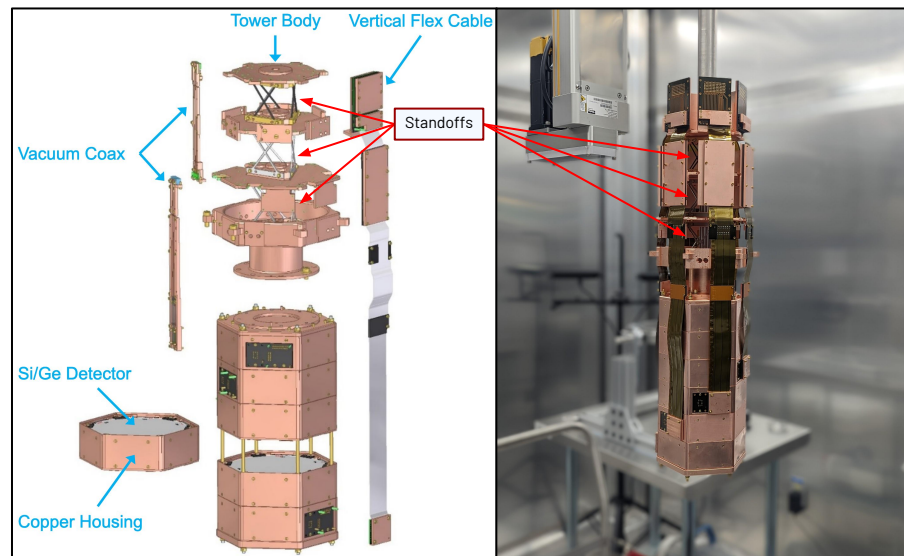
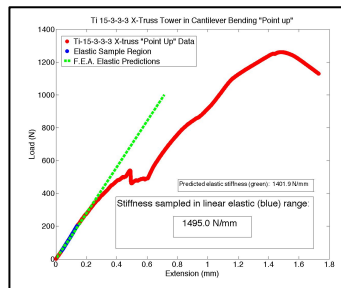
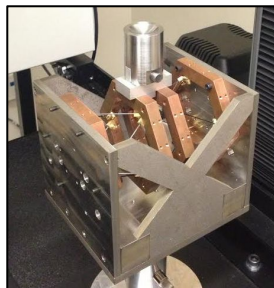
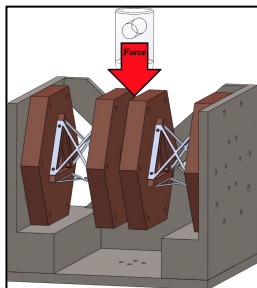
Erik Kramer | SuperCDMS SNOLAB Support Tower Standoffs

Project → A multi-temperature stage support structure for cryogenic dark matter detectors

Tasks → **Design**, **delivery** and **qualification** of structural thermal standoffs for multiple temperature stages



Design Trades, Analysis, and Testing



Final Design Integration

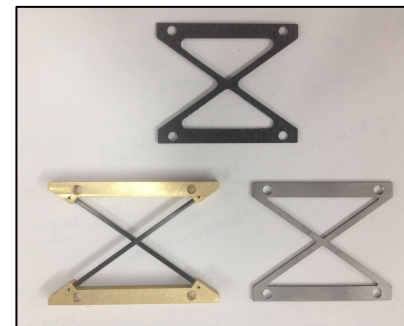
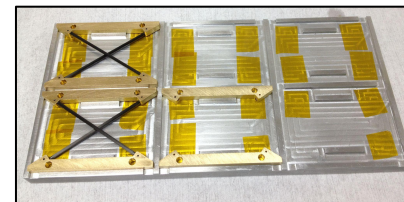
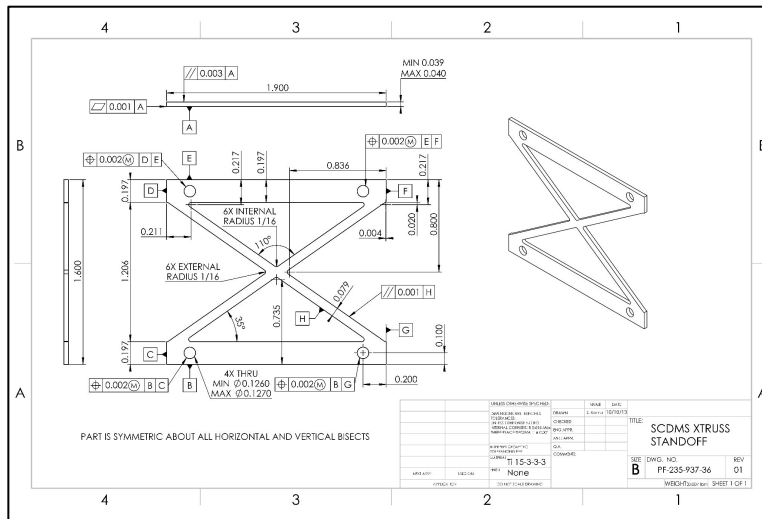
Objectives

- Design standoffs to structurally support multiple temperature stages for the SuperCDMS SNOLAB experiment tower
- Fabricate hardware and qualify based on structural and thermal requirements

- Design standoffs to structurally support multiple temperature stages for the SuperCDMS SNOLAB experiment tower
- Fabricate hardware and qualify based on structural and thermal requirements

- Used **SolidWorks** and **MATLAB** to design and model both **structural** and **thermal** properties
- Produced ASME Y14.5 **GD&T** standard drawings and oversaw fabrication of hardware and test equipment
- Performed **qualification testing**

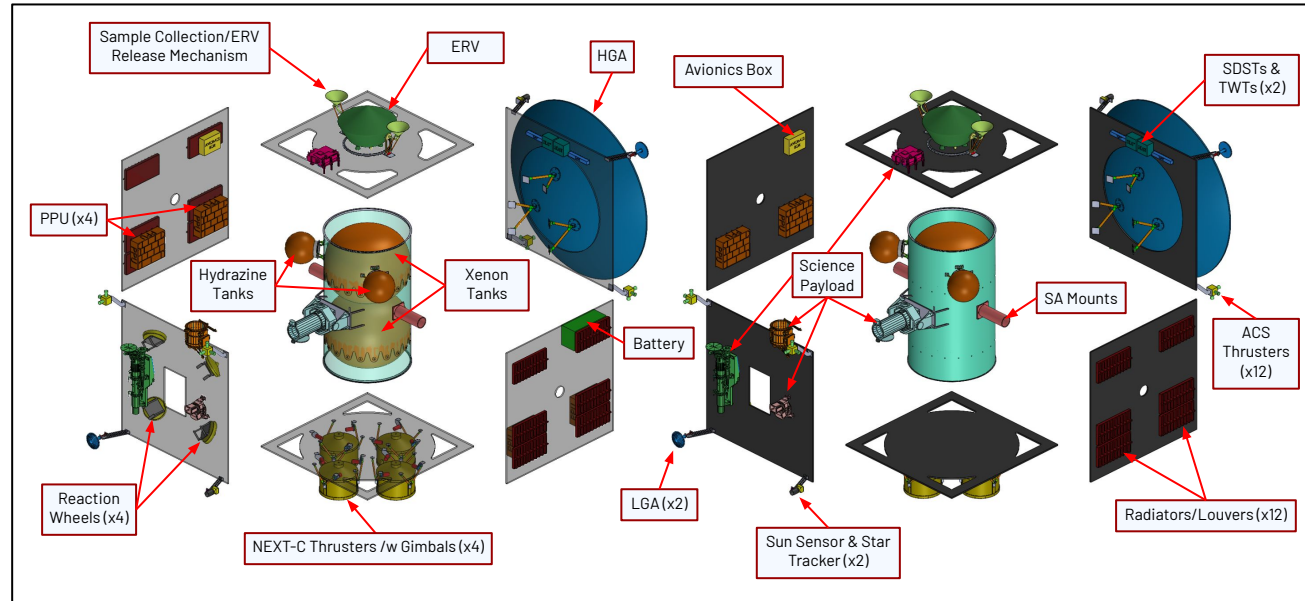
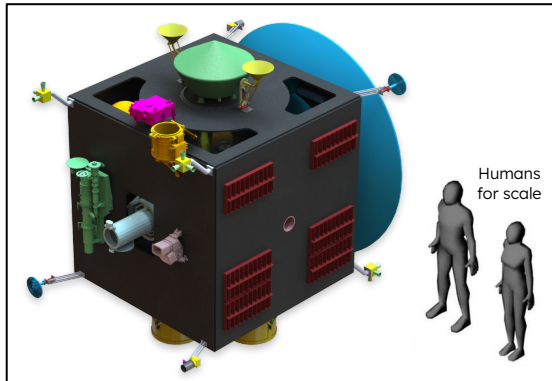
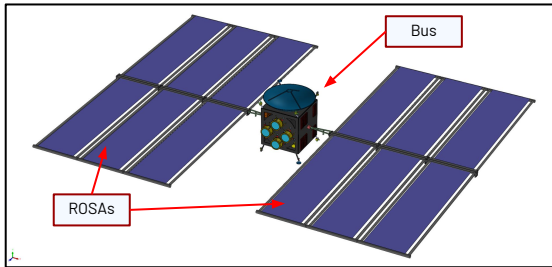
- Timely delivery of 2 final hardware designs that meet requirements
- Standoffs are now in active use as part of the SuperCDMS SNOLAB experiment



Erik Kramer | Europa Sample Return Mission Concept Study

Project → A pre-Phase A type concept study for a Europa sample return mission

Tasks → Design of **structures** subsystem, **CAD** of all spacecraft components/configurations, **systems engineering** work



Spacecraft Design and Configuration

Erik Kramer | Europa Sample Return Mission Concept Study

Objectives

- Perform a concept study for a mission to return a water plume sample from Europa
- Create a proposal with details for all subsystems showing mission viability
- Launch within 2030-2033 and have a mission duration of 10-12 years

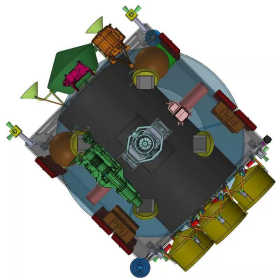
Process

- Used **SolidWorks** to design and model 100+ spacecraft components
- Tracked parts list, mass, and power through linked design spreadsheets, **MEL** and **PEL**
- Assisted other subsystems with **design iterations** and **trade studies**

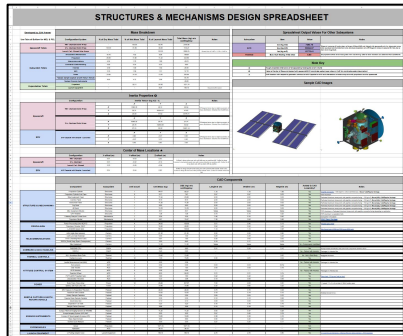
Results

- Produced a comprehensive SEP mission concept, including spacecraft design, that satisfies all level 1 to 3 requirements and lasts 11.14 years
- Collaborated to produce a 130 page report documenting the design and mission proposal

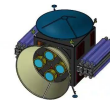
[Click](#) for Transparent Bus Rotation Animation



[Click](#) for Design Spreadsheet Example

A screenshot of a complex spreadsheet titled "STRUCTURES & MECHANISMS DESIGN SPREADSHEET". It contains multiple tabs and a large grid of data. The visible data includes columns for item numbers, descriptions, materials, and various engineering parameters. There are also smaller diagrams and charts embedded within the spreadsheet, such as a small 3D model of a component and a graph showing a relationship between variables.

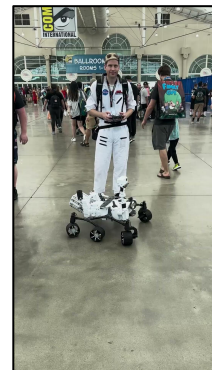
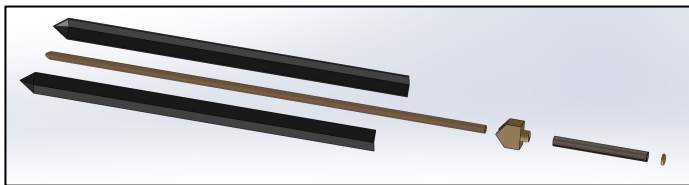
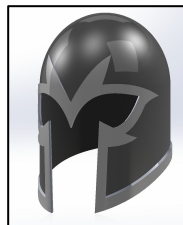
[Click](#) for Solar Array Deployment Animation



Erik Kramer | Just For Fun

Project → 50+ costumes and props, including mechatronic and articulating items, for comic conventions and contests

Tasks → Translation of images and concepts into real world products through **engineering approaches**



[Click for Rover Backpack Video](#)

